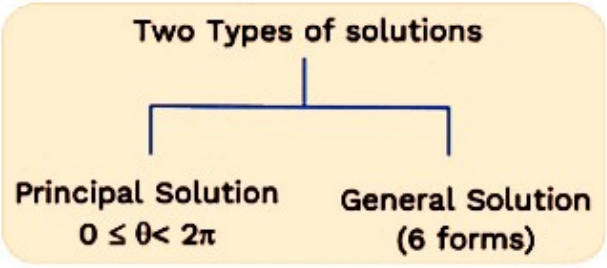
Trigonometric Equations

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Trigonometric Equations



General Solutions of Trigonometric Equation:

(i) If $\sin\theta = \sin\alpha$

$\Rightarrow$ $\theta = n\pi + (-1)^n\alpha$

Where $\alpha \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$, $n \in I$

(ii) If $\cos\theta = \cos\alpha$

$\Rightarrow$ $\theta = 2n\pi \pm \alpha$

Where $\alpha \in [0, \pi]$, $n \in I$

(iii) If $\tan\theta = \tan\alpha$

$\Rightarrow$ $\theta = n\pi + \alpha$

Where $\alpha \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, $n \in I$

(iv) If $\sin^2\theta = \sin^2\alpha$

$\Rightarrow$ $\theta = n\pi \pm \alpha$

Where $\alpha \in \left[0, \frac{\pi}{2}\right]$, $n \in I$

(v) If $\cos^2\theta = \cos^2\alpha$

$\Rightarrow$ $\theta = n\pi \pm \alpha$

Where $\alpha \in \left[0, \frac{\pi}{2}\right]$, $n \in I$

(vi) If $\tan^2\theta = \tan^2\alpha$

$\Rightarrow$ $\theta = n\pi \pm \alpha$

Where $\alpha \in \left[0, \frac{\pi}{2}\right)$, $n \in I$

Q.1 What is the most general values of θ which satisfy both the equations.

$\sin\theta = -\frac{1}{2}$ and $\tan\theta = \frac{1}{\sqrt{3}}$

A. Clearly θ lies in III Quadrant.

Hence common values for both equations are

$$\theta = (2n+1)\pi + \frac{\pi}{6}$$

or $\theta = 2n\pi + \frac{7\pi}{6}, n \in I$

Q.2 Principle solutions of equation : $\left(\sin x + \frac{1}{2}\right)^2 + \left(\tan x + 1/\sqrt{3}\right)^2 = 0$

A. Clearly $\sin x = -\frac{1}{2}$ and $\tan x = -\frac{1}{\sqrt{3}}$

$\Rightarrow$ $\theta \in$ IV Quadrant

Hence $\theta = 2n\pi - \frac{\pi}{6}$, $n \in I$

Principal solution $(0 \le \theta < \pi)$

Put $n = 1$ $\theta = \frac{11\pi}{6}$

Types of Trigonometric Equations

(i) Type – 1

Solution of trigonometric equation by factorization or equation which are expressed in quadratic form.

Solve the following trigonometric equations:

Q.1 $2\cos x.\cos 2x = \cos x$

A. $2\cos x.\cos 2x - \cos x = 0$

$\cos x(2\cos 2x - 1) = 0$

$\cos x = 0$ or $\cos 2x = \frac{1}{2}$

$x = (2n+1)\frac{\pi}{2}$ $(n \in I)$ $\cos 2x = \cos\frac{\pi}{3}$

$2x = 2n\pi \pm \frac{\pi}{3}$

$x = n\pi \pm \frac{\pi}{6}$ $(n \in I)$

Q.2 $\cot x - \cos x = 1 - \cot x.\cos x$

A.

$\cot x - 1 - \cos x + \cot x.\cos x = 0$

$(\cot x - 1) + \cos x(-1 + \cot x) = 0$

$(\cot x - 1)(\cos x + 1) = 0$

$\cot x = 1$	or	$\cos x = -1$
$\tan x = \tan\frac{\pi}{4}$		$x = (2n + 1)\pi$ (cot x is not defined)
		so, rejected ($n \in I$)

$x = n\pi + \frac{\pi}{4}$ $(n \in I)$

Q.3 $3\cos^2 x - 10\cos x + 3 = 0$

A.

$3\cos^2 x - 9\cos x - \cos x + 3 = 0$

$3\cos x(\cos x - 3) - (\cos x - 3) = 0$

$(3\cos x - 1)(\cos x - 3) = 0$

$\cos x = 1/3$	or	$\cos x = 3$
$\cos x = \cos\alpha$		$x \in \phi$
$x = 2n\pi \pm \alpha$		

$(n \in I, \cos\alpha = \frac{1}{3})$

Q.4 $3\sin^2 x - \sin x \cos x - 4\cos^2 x = 0$

A.

$3\sin^2 x - 4\sin x \cos x + 3\sin x \cos x - 4\cos^2 x = 0$

$\sin x(3\sin x - 4\cos x) + \cos x(3\sin x - 4\cos x) = 0$

$(\sin x + \cos x)(3\sin x - 4\cos x) = 0$

$\sin x = -\cos x$	or	$3\sin x = 4\cos x$
$\tan x = -1$		$\tan x = 4/3$
$\tan x = \tan(-\pi/4)$		$\tan x = \tan\alpha$
$x = n\pi + (-\pi/4)$		$x = n\pi + \alpha$
$(n \in I)$		$(n \in I, \tan\alpha = \frac{4}{3})$

Q.5 $2\sin^2 2x + 6\sin^2 x = 5$

A.

$2(1-\cos^2 2x) + 6\sin^2 x = 5$

$2 - 2\cos^2 2x + 6\sin^2 x = 5$

$2 - 2(1 - 2\sin^2 x)^2 + 6\sin^2 x = 5$

Let $\sin^2 x = a$ then

$2 - 2(1 - 2a)^2 + 6a = 5$

$2 - 2(1 - 4a + 4a^2) + 6a = 5$

$8a^2 - 14a + 5 = 0$

$8a^2 - 10a - 4a + 5 = 0$

$2a(4a - 5) - (4a - 5) = 0$

$(2a - 1)(4a - 5) = 0$

$\Rightarrow a = ½$ or $a = 5/4$

$\sin^2 x = ½$ $\sin^2 x = 5/4$ (not possible)

$\sin^2 x = \sin^2 \pi/4$

$x = \mathbf{n\pi \pm \pi/4}$

$(n \in I)$

Q.6 $(1 - \tan\theta)(1 + \sin 2\theta) = (1 + \tan\theta)$

A. $(1 - \tan\theta)\left(1 + \frac{2\tan\theta}{1 + \tan^2\theta}\right) = (1 + \tan\theta)$

$(1 - \tan\theta)\frac{(1 + \tan\theta)^2}{1 + \tan^2\theta} = (1 + \tan\theta)$

$(1 - \tan\theta)(1 + \tan\theta)^2 - (1 + \tan\theta)(1 + \tan^2\theta) = 0$

$(1 + \tan\theta)\{(1 - \tan\theta)(1 + \tan\theta) - (1 + \tan^2\theta)\} = 0$

$(1 + \tan\theta)\{(1 - \tan^2\theta) - (1 + \tan^2\theta)\} = 0$

$\tan\theta = -1$ or $\tan\theta = 0$

$\tan\theta = \tan(-\pi/4)$ $\theta = n\pi$ $(n \in I)$

$\theta = \mathbf{n\pi - \frac{\pi}{4}}, (n \in I)$

Q.7 Find the general solution of the trigonometric equation $\cos 4x + 6 = 7\cos 2x$ and also find sum of all the solution in $[0, 100\pi]$

A. $\cos 4x + 6 = 7\cos 2x$

$(2\cos^2 2x - 1) + 6 = 7\cos 2x$

$2\cos^2 2x - 7\cos 2x + 5 = 0$

$2\cos^2 2x - 2\cos 2x - 5\cos 2x + 5 = 0$

$2\cos 2x(\cos 2x - 1) - 5(\cos 2x - 1) = 0$

$\cos 2x = 5/2$ or $\cos 2x = 1$

(not possible) $2x = 2n\pi$

$x = n\pi, (n \in I)$

Sum of all solutions $= 0 + \pi + 2\pi + \ldots + 100\pi$

$= \pi\left(\frac{100.101}{2}\right) = \mathbf{5050\pi}$

Previous Year's Questions

Let $S = \{\theta \in [-2\pi, 2\pi] : 2\cos^2\theta + 3\sin\theta = 0\}$. Then the sum of the elements of S is

(A) $\frac{13\pi}{6}$ (B) π

(C) 2π (D) $\frac{5\pi}{6}$

Q.8 $5\tan^4x - (\sec^2x)^2 = 29$

A. $5(\tan^2x)^2 - (1 + \tan^2x)^2 = 29$

let $\tan^2x = a$ then

$5a^2 - (1 + a)^2 = 29$

$5a^2 - (1 + a^2 + 2a) = 29$

$4a^2 - 2a - 30 = 0$

$2a^2 - a - 15 = 0 \Rightarrow 2a^2 - 6a + 5a - 15 = 0$

$2a(a - 3) + 5(a - 3) = 0$

$(2a + 5)(a - 3) = 0$

$a = -\frac{5}{2}$ or $a = 3$

$\tan^2x = -\frac{5}{2}$ $\qquad$ $\tan^2x = \tan^2\frac{\pi}{3}$

(Not possible) $\qquad$ $x = n\pi \pm \frac{\pi}{3}, \quad n \in I$

Q.9 $\sin^2x(1 + \tan x) = 3\sin x(\cos x - \sin x) + 3$

A. $\sin^2x(1 + \tan x) = 3\sin x \cos x(1 - \tan x) + 3$

divide by $\cos^2x$

$\tan^2x(1 + \tan x) = 3\tan x(1 - \tan x) + 3\sec^2x$

$\tan^2x + \tan^3x = 3\tan x - 3\tan^2x + 3 + 3\tan^2x$

$\tan^3x + \tan^2x - 3\tan x - 3 = 0$

$\tan^2x(\tan x + 1) - 3(\tan x + 1) = 0$

$(\tan^2x - 3)(\tan x + 1) = 0$

$\tan^2x = 3$ or $\tan x = -1$

$\tan^2x = \tan^2\frac{\pi}{3}$ $\qquad$ $\tan x = \tan\left(-\frac{\pi}{4}\right)$

$x = n\pi \pm \frac{\pi}{3}$ $\qquad$ $x = n\pi - \frac{\pi}{4}$

$(n \in I)$ $\qquad$ $(n \in I)$

Q.10 Find the solutions set of the equation

$$\log_{\frac{-x^2-6x}{10}}(\sin 3x + \sin x) = \log_{\frac{-x^2-6x}{10}}(\sin 2x)$$

A. For log to be defined

(i) $\sin 3x + \sin x > 0$

(ii) $\sin 2x > 0$

(iii) $\frac{-x^2 - 6x}{10} > 0 \Rightarrow x(x + 6) < 0$

(iv) $-\frac{x^2-6x}{10} \neq 1 \Rightarrow x^2+6x+10 \neq 0$

Now from given equation

$\Rightarrow \sin3x + \sin x = \sin2x$

$\Rightarrow 2\sin2x\cos x - \sin2x = 0$

$\Rightarrow \sin2x(2\cos x - 1) = 0$

$\Rightarrow \sin2x = 0$ or $\cos x = \frac{1}{2}$

$(\because \sin2x > 0)$ $x = 2n\pi \pm \pi/3$

rejected $(n \in I)$

$\Rightarrow x = -2\pi + \frac{\pi}{3}, -\frac{\pi}{3}$ $\{\because x \in (-6, 0)\}$

$x = -\frac{5\pi}{3}$ is only a solution.

Q.11 $\tan^2\theta + \sec2\theta = 1$

A. $\tan^2\theta + \frac{1}{\cos2\theta} = 1$

$\tan^2\theta + \frac{1+\tan^2\theta}{1-\tan^2\theta} = 1$

Let $\tan^2\theta = a$ then

$a + \frac{1+a}{1-a} = 1 \Rightarrow a(1-a) + (1+a) = (1-a)$

$\Rightarrow a - a^2 + 1 + a = 1 - a$

$\Rightarrow a^2 - 3a = 0$

$\Rightarrow a(a-3) = 0$

$a = 0$ or $a = 3$

$\tan^2\theta = 0$ or $\tan^2\theta = \tan^2\pi/3$

$\theta = n\pi, (n \in I)$ $\theta = n\pi \pm \pi/3$ $(n \in I)$

(ii) Type – 2

Solving trigonometric equation by introducing an Auxiliary argument

equation of the form

$\Rightarrow \quad a\cos\theta + b\sin\theta = c$

divide by $\sqrt{a^2+b^2}$

$\Rightarrow \underbrace{\frac{a}{\sqrt{a^2+b^2}}}_{\sin\alpha}\cos\theta + \underbrace{\frac{b}{\sqrt{a^2+b^2}}}_{\cos\alpha}\sin\theta = \frac{c}{\sqrt{a^2+b^2}}$

$\Rightarrow \sin(\theta+\alpha) = \frac{c}{\sqrt{a^2+b^2}}$

Point to Remember!!!

(i) First check for real solution

(ii) Avoid squaring

Q.1 Solve: $\sin x + \cos x = \sqrt{2}$

A. Divide by $\sqrt{1+1} = \sqrt{2}$

$\Rightarrow \frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x = \frac{\sqrt{2}}{\sqrt{2}}$

$\Rightarrow \cos\frac{\pi}{4} \sin x + \sin\frac{\pi}{4} \cos x = 1$

$\Rightarrow \sin(x + \pi/4) = 1$

$\Rightarrow x + \frac{\pi}{4} = 2n\pi + \frac{\pi}{2}$

$\Rightarrow x = \mathbf{2n\pi + \frac{\pi}{4}}$ $(n \in I)$

Q.2 Solve: $\sqrt{3}\cos x + \sin x = 2$

A. Divide by $\sqrt{3+1} = 2$

$\Rightarrow \frac{\sqrt{3}}{2} \cos x + \frac{1}{2} \sin x = \frac{2}{2}$

$\Rightarrow \sin\frac{\pi}{3} \cos x + \cos\frac{\pi}{3} \sin x = 1$

$\Rightarrow \sin(x + \quad /3) = 1$

$\Rightarrow x + \frac{\pi}{3} = 2n\pi + \frac{\pi}{2}$

$\Rightarrow x = \mathbf{2n\pi + \frac{\pi}{6}}$ $(n \in I)$

Q.3 Solve: $\sin x + \cos x = 1.5$

A. Divide by $\sqrt{1+1} = \sqrt{2}$

$\Rightarrow \frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x = \frac{3}{2\sqrt{2}}$

$\Rightarrow \sin(x + \frac{\pi}{4}) = \frac{3}{2\sqrt{2}}$

$\Rightarrow$ **No solution** $(\because \frac{3}{2\sqrt{2}} > 1)$

Q.4 Solve: $(\sec x - 1) = (\sqrt{2} - 1)\tan x$

A. $\left(\frac{1}{\cos x} - 1\right) = (\sqrt{2} - 1)\frac{\sin x}{\cos x}$

$\Rightarrow 1 - \cos x = (\sqrt{2} - 1)\sin x;\ \cos x \neq 0$

$\Rightarrow 2\sin^2\frac{x}{2} = (\sqrt{2} - 1)\, 2\sin\frac{x}{2} \cdot \cos\frac{x}{2}$

$\Rightarrow 2\sin\frac{x}{2} = 0$ or $\sin\frac{x}{2} = (\sqrt{2} - 1)\cos\frac{x}{2}$

$\Rightarrow \frac{x}{2} = n\pi\ (n \in I)$ $\tan\frac{x}{2} = \sqrt{2} - 1 = \tan\frac{\pi}{8}$

$\Rightarrow x = \mathbf{2n\pi}\ (n \in I)$ $\Rightarrow \frac{x}{2} = n\pi + \frac{\pi}{8}(n \in I)$

$\Rightarrow x = \mathbf{2n\pi} + \frac{\pi}{4}\ (n \in I)$

Q.5 Solve: $\sin x + \cos x = \sqrt{2}\ \sin 5x$

A. Divide by $\sqrt{1+1} = \sqrt{2}$

$\Rightarrow \sin(x + \frac{\pi}{4}) = \sin 5x$

$\Rightarrow x + \frac{\pi}{4} = n\pi + (-1)^n 5x$ $(n \in I)$

for n → even let n = 2m

$\Rightarrow x + \frac{\pi}{4} = 2m\pi + 5x$

$\Rightarrow 4x = -2m\pi + \frac{\pi}{4}$

$\Rightarrow x = \frac{-m\pi}{2} + \frac{\pi}{16}$

or $x = k\frac{\pi}{2} + \frac{\pi}{16}(k \in I)$

for n → odd, let n = (2m + 1)

$x + \frac{\pi}{4} = (2m + 1)\pi - 5x$

$\Rightarrow 6x = 2m\pi + \frac{3\pi}{4}$

$\Rightarrow x = \frac{m\pi}{3} + \frac{\pi}{8}\ (m \in I)$

Q.6 Solve the equality: $2\sin 11x + \cos 3x + \sqrt{3}\sin 3x = 0$

A. $\cos 3x + \sqrt{3}\sin 3x = -2\sin 11x$

Divide by $\sqrt{1+3} = 2$

$$\Rightarrow \frac{1}{2}\cos 3x + \frac{\sqrt{3}}{2}\sin 3x = -\sin 11x$$

$$\Rightarrow \sin\left(3x + \frac{\pi}{6}\right) = -\sin 11x$$

$$\Rightarrow \left(3x + \frac{\pi}{6}\right) = n\pi + (-1)^n(-11x)$$

If $n \to$ even $\Rightarrow n = 2m$

$$\Rightarrow 3x + \frac{\pi}{6} = 2m\pi - 11x$$

$$\Rightarrow 14x = 2m\pi - \frac{\pi}{6}$$

$$\Rightarrow x = \frac{m\pi}{7} - \frac{\pi}{84} \quad (m \in I)$$

If $n \to$ odd $\Rightarrow n = 2m + 1$

$$\Rightarrow 3x + \frac{\pi}{6} = (2m+1)\pi + 11x$$

$$\Rightarrow 8x = -2m\pi - \frac{5\pi}{6}$$

$$\Rightarrow x = -\frac{2m\pi}{8} - \frac{5\pi}{48}$$

or $x = \frac{k\pi}{4} - \frac{5\pi}{48}$ $(k \in I)$

Q.7 If α and β are the roots of the equation $a\cos\theta + b\sin\theta = c$ then match the entries of column -I with the entries of column – II

Column – I	**Column – II**
(A) $\sin\alpha + \sin\beta$	**(P)** $2b/(a+c)$
(B) $\sin\alpha \cdot \sin\beta$	**(Q)** $(c-a)/(c+a)$
(C) $\tan\frac{\alpha}{2} + \tan\frac{\beta}{2}$	**(R)** $2bc/(a^2+b^2)$
(D) $\tan\frac{\alpha}{2} \cdot \tan\frac{\beta}{2}$	**(S)** $(c^2-a^2)/(a^2+b^2)$

A. **A → R, B → S, C → P, D → Q**

$a\cos\theta = c - b\sin\theta$

on squaring

$\Rightarrow a^2\cos^2\theta = c^2 + b^2\sin^2\theta - 2bc\sin\theta$

$\Rightarrow a^2(1 - \sin^2\theta) = c^2 + b^2\sin^2\theta - 2bc\sin\theta$

Let $\sin\theta = x$

$\Rightarrow (a^2 + b^2)x^2 - 2bc.\ x + (c^2 - a^2) = 0$

above equation have roots $\sin\alpha$, $\sin\beta$,

$$\Rightarrow \sin\alpha + \sin\beta = \frac{2bc}{a^2+b^2}$$

$$\Rightarrow \sin\alpha . \sin\beta = \frac{c^2-a^2}{a^2+b^2}$$

Now from given equation

$$a\left(\frac{1-\tan^2\left(\frac{\theta}{2}\right)}{1+\tan^2\left(\frac{\theta}{2}\right)}\right)+b\left(\frac{2\tan\left(\frac{\theta}{2}\right)}{1+\tan^2\left(\frac{\theta}{2}\right)}\right)=c$$

$\Rightarrow a(1 - \tan^2\theta/2) + 2b\tan\theta/2 = c(1 + \tan^2\theta/2)$

Let $\tan\theta/2 = x$

$\Rightarrow (a + c)x^2 - 2bx + (c - a) = 0$

above equation have roots $\tan\alpha/2$, $\tan\beta/2$

$\Rightarrow \tan\alpha/2 + \tan\beta/2 = 2b/(a+c)$

$$\Rightarrow \tan\frac{\alpha}{2} . \tan\frac{\beta}{2} = (c - a)/(c + a)$$

Q.8 Solve: $4\cos x + 3\sin x = 5$

A. Divide by $\sqrt{4^2 + 3^2} = 5$

$$\Rightarrow \frac{4}{5}\cos x + \frac{3}{5}\sin x = 1$$

$$\Rightarrow \sin(x + \alpha) = 1 \text{ where } \sin\alpha = \frac{4}{5}$$

$$\Rightarrow x + \alpha = 2n\pi + \frac{\pi}{2} \ (n \in I)$$

$$\Rightarrow x = \mathbf{2n\pi + \frac{\pi}{2} - \alpha} \ (n \in I, \mathbf{\sin\alpha = \frac{4}{5}})$$

Q.9 Solve: $\sin\left(\frac{\sqrt{x}}{2}\right) + \cos\left(\frac{\sqrt{x}}{2}\right) = \sqrt{2}\ \sin\sqrt{x}$

A. Divide by $\sqrt{2}$

$$\sin\left(\frac{\sqrt{x}}{2} + \frac{\pi}{4}\right) = \sin(\sqrt{x})$$

$$\Rightarrow \frac{\sqrt{x}}{2} + \frac{\pi}{4} = n\pi + (-1)^n \sqrt{x} \quad (n \in I, x \geq 0)$$

If $n = -2m$ (even) then

$$\frac{\sqrt{x}}{2} + \frac{\pi}{4} = -2m + \sqrt{x}$$

$$\Rightarrow \frac{\sqrt{x}}{2} = 2m\pi + \frac{\pi}{4}$$

$$\Rightarrow x = \left(4m\pi + \frac{\pi}{2}\right)^2 \quad (m \in I)$$

If $n = 2m + 1$ (odd) then

$$\frac{\sqrt{x}}{2} + \frac{\pi}{4} = (2m + 1)\pi - \sqrt{x}$$

$$\Rightarrow \frac{3\sqrt{x}}{2} = 2m\pi + \frac{3\pi}{4}$$

$$\Rightarrow x = \left(\frac{4m\pi}{3} + \frac{\pi}{2}\right)^2 \quad (m \in I)$$

Q.10 Solve: $1 + \sin^3 x + \cos^3 x = \frac{3}{2}\sin 2x$

A. Given equation is

$1^3 + (\sin x)^3 + (\cos x)^3 = 3(1)(\sin x)(\cos x)$

$\Rightarrow 1 + \sin x + \cos x = 0$ or $1 = \sin x = \cos x$ (Not possible)

$\Rightarrow \sin x + \cos x = -1$

$$\Rightarrow \sin\left(x + \frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}} = \sin\left(-\frac{\pi}{4}\right)$$

$$\Rightarrow x + \frac{\pi}{4} = n\pi + (-1)^n\left(-\frac{\pi}{4}\right)$$

If $n = 2m$ (even)

$$\Rightarrow x + \frac{\pi}{4} = 2m\pi - \frac{\pi}{4}$$

$$\Rightarrow x = 2m\pi - \frac{\pi}{2} \quad (m \in I)$$

If n = 2m + 1 (odd)

$\Rightarrow$ $x + \pi/4 = (2m + 1)\pi + \pi/4$

$\Rightarrow$ $x = \mathbf{(2m + 1)\pi}$ $(m \in I)$

Q.11 Number of integral values of k for which the equation $3\sin x + 4\cos x = k$ has real solutions.

A. $\because -\sqrt{3^2+4^2} \le 3\sin x + 4\cos x \le \sqrt{3^2+4^2}$

$\therefore -5 \le k \le 5$

Integral values of k are

{0, ±1, ±2, ±3, ±4, ±5}

Number of values of k = **11**

Q.12 The number of integral values of k for which the equation $7\cos x + 5\sin x = 2k + 1$ has a solution

(A) 4 (B) 8 (C) 10 (D) 12

A. **B**

We know that

$-\sqrt{7^2+5^2} \le 7\cos x + 5\sin x \le \sqrt{7^2+5^2}$

$-\sqrt{74} \le 2k + 1 \le \sqrt{74}$

Integral values of k are

{0, ±1, ±2, ±3, -4}

Number of values = **8**

Q.13 Find the general solution of the trigonometric equation

$3^{\left\{\frac{1}{2}+\log_2(\cos x+\sin x)\right\}} - 2^{\log_2(\cos x-\sin x)} = \sqrt{2}$

A. Given equation is

$\sqrt{3}\cdot 3^{\log_2(\cos x+\sin x)} - 2^{\log_2(\cos x-\sin x)} = \sqrt{2}$

$\Rightarrow \sqrt{3}(\cos x+\sin x) - (\cos x-\sin x) = \sqrt{2}$

Divide by $\sqrt{2}$

$$\Rightarrow \sqrt{3}\sin\left(x+\frac{\pi}{4}\right) - \cos\left(x+\frac{\pi}{4}\right) = 1$$

Divide by 2

$$\frac{\sqrt{3}}{2}\sin\left(x+\frac{\pi}{4}\right) - \frac{1}{2}\cos\left(x+\frac{\pi}{4}\right) = \frac{1}{2}$$

$\Rightarrow \sin(x + \frac{\pi}{4} - \frac{\pi}{6}) = \sin\frac{\pi}{6}$

$\Rightarrow \quad x + \frac{\pi}{12} = n\pi + (-1)^n \frac{\pi}{6} \; (n \in I)$

If $n = 2m$ (even) then

$\Rightarrow \quad x = 2m\pi + \frac{\pi}{12} \; (m \in I)$

If $n = 2m + 1$ (odd) then

$\Rightarrow \quad x = \mathbf{(2m + 1)\pi - \frac{\pi}{4}} \; (m \in I)$ (rejected)

(as $\sin x + \cos x = 0$)

Q.14 Solve: $3\sqrt{3}\sin^3 x + \cos^3 x + 3\sqrt{3}\sin x \cos x = 1$

A. Given equation is

$(\sqrt{3}\sin x)^3 + (\cos x)^3 + (-1)^3 = 3(\sqrt{3}\sin x)\cos x.(-1)$

$\Rightarrow \sqrt{3}\sin x + \cos x - 1 = 0$ or $\sqrt{3}\sin x = \cos x = -1$

$\frac{\sqrt{3}}{2}\sin x + \frac{1}{2}\cos x = \frac{1}{2}$

$\cos\left(x - \frac{\pi}{3}\right) = \cos\frac{\pi}{3}$

$\Rightarrow x - \frac{\pi}{3} = 2n\pi \pm \frac{\pi}{3} \; (n \in I)$

$\Rightarrow \mathbf{x = 2n\pi + \frac{2\pi}{3}}$ or $\mathbf{x = 2n\pi} \; (n \in I)$

(iii) Type – 3

Solving equations by transforming a sum of Trigonometric functions into a product.

Q.1 General solution of the trigonometric equations $\sin x + \sin 5x = \sin 2x + \sin 4x$ is

(A) $\frac{n\pi}{3}$ (B) $\frac{2n\pi}{3}$ (C) $2n\pi$ (D) $n\pi$

A. **A**

$2\sin 3x.\cos 2x = 2\sin 3x.\cos x$

$\Rightarrow \sin 3x = 0$ or $\cos 2x = \cos x$

$3x = n\pi$ $\quad 2x = 2n\pi \pm x$

$x = \frac{n\pi}{3}(n \in I)$ $\quad x = 2n\pi, \frac{2n\pi}{3} \; (n \in I)$

Combinedly $\mathbf{x = \frac{n\pi}{3}}$

(iv) Type – 4

Solving equations by transforming a product of trigonometric functions into a sum.

Q.1 Number of solutions of the trigonometric equation in $[0, \pi]$,

$\sin 3\theta = 4\sin\theta \sin 2\theta \sin 4\theta$

(A) 4 (B) 6 (C) 7 (D) 10

A. **C**

$\sin 3\theta = 4\sin\theta \sin 2\theta \sin 4\theta$

$\Rightarrow \sin 3\theta = 2(\cos\theta - \cos 3\theta)\sin 4\theta$

$\Rightarrow \sin 3\theta = 2\sin 4\theta.\cos\theta - 2\sin 4\theta\cos 3\theta$

$\Rightarrow \sin 3\theta = (\sin 5\theta + \sin 3\theta) - (\sin 7\theta + \sin\theta)$

$\Rightarrow (\sin 7\theta - \sin 5\theta) + \sin\theta = 0$

$\Rightarrow 2\cos 6\theta.\sin\theta + \sin\theta = 0$

$\Rightarrow \sin\theta(2\cos 6\theta + 1) = 0$

$\sin\theta = 0$ or $\cos 6\theta = -\frac{1}{2}$

$\theta = n\pi$ $(n \in I)$ $6\theta = 2n\pi \pm \frac{2\pi}{3}$

Values in $[0, \pi] \Rightarrow 0, \pi, \frac{8\pi}{18}, \frac{4\pi}{18}, \frac{14\pi}{18}, \frac{10\pi}{18}, \frac{16\pi}{18}$

Q.2 Find all value of θ, between 0 and π, which satisfy the equation

$\cos\theta.\cos 2\theta.\cos 3\theta = \frac{1}{4}$

A. $2(\cos\theta.\cos 3\theta).\cos 2\theta = \frac{1}{2}$

$\Rightarrow (\cos 4\theta + \cos 2\theta).\cos 2\theta = \frac{1}{2}$

$\Rightarrow (2\cos^2 2\theta - 1 + \cos 2\theta).\cos 2\theta = \frac{1}{2}$

let $\cos 2\theta = x$ Hence

$\Rightarrow (2x^2 + x - 1).x = \frac{1}{2}$

$\Rightarrow 4x^3 + 2x^2 - 2x - 1 = 0$

$\Rightarrow 2x^2(2x + 1) - (2x + 1) = 0$

$\Rightarrow \quad (2x^2 - 1)(2x + 1) = 0$

$\Rightarrow \quad x^2 = ½ \qquad$ or $\qquad x = -1/2$

$\cos^2 2\theta = \cos^2 \frac{\pi}{4} \qquad \cos 2\theta = \cos \frac{2\pi}{3}$

$\theta = \frac{n\pi}{2} \pm \frac{\pi}{8} (n \in I) \qquad \theta = n\pi \pm \frac{\pi}{3} (n \in I)$

In $[0, \pi]$ values are $\theta = \frac{\pi}{8}, \frac{\pi}{3}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{2\pi}{3}$

Q.3 Number of solution(s) of the equation $\sin x + 2\sin 2x = 3 + \sin 3x$ in $[0, \pi]$

(A) no solution (B) infinite solution

(C) exactly one solution (D) exactly two solution

A. **A**

$\sin 3x - 2\sin 2x - \sin x = -3$

$3\sin x - 4\sin^3 x - 4\sin x \cos x - \sin x = -3$

$2\sin x - 4\sin^3 x - 4\sin x \cos x = -3$

$2\sin x(1 - 2\sin^2 x - 2\cos x) = -3$

$2\cos^2 x - 2\cos x - 1 = -\frac{3}{2}\operatorname{cosec} x$

$2\left(\cos^2 x - \cos x - \frac{1}{2}\right) = -\frac{3}{2}\operatorname{cosec} x$

$2\left\{\left(\cos x - \frac{1}{2}\right)^2 - \frac{3}{4}\right\} = -\frac{3}{2}\operatorname{cosec} x$

for $x \in [0, \pi] \qquad LHS_{min} = -\frac{3}{2}$

$RHS_{max} = -\frac{3}{2}$

Hence LHS = RHS $= -\frac{3}{2}$

$\Rightarrow \cos x = -\frac{1}{2}$ and $\operatorname{cosec} x = 1$

$\cos x = -\frac{1}{2}$ and $\sin x = 1$ (Not possible simultaneously)

$\Rightarrow$ **No Solutions**

Alternate method:

$(\sin 3x - \sin x) - 2\sin 2x = 3$

$2\cos 2x \sin x - 2\sin 2x = 3$

$(\sin x)\cos 2x - \sin 2x = -\frac{3}{2}$

LHS $\in \left[-\sqrt{1+\sin^2 x}, \sqrt{1+\sin^2 x}\right]$

which cannot be $\frac{3}{2}$ hence **no solutions.**

Q.4 Solve: $\cos^2 x + \cos^2 2x + \cos^3 3x + \cos^2 4x = 2$

A. $\left(\frac{1+\cos 2x}{2}\right)+\left(\frac{1+\cos 4x}{2}\right)+\left(\frac{1+\cos 6x}{2}\right)+\left(\frac{1+\cos 8x}{2}\right) = 2$

$\Rightarrow \frac{1}{2}(\cos 2x + \cos 4x + \cos 6x + \cos 8x) = 0$

$\Rightarrow (\cos 2x + \cos 8x) + (\cos 4x + \cos 6x) = 0$

$\Rightarrow 2\cos 5x \cos 3x + 2\cos 5x.\ \cos x = 0$

$\Rightarrow 2\cos 5x(\cos 3x + \cos x) = 0$

$\Rightarrow 2\cos 5x\ (2\cos 2x.\ \cos x) = 0$

$\cos 5x = 0$	or	$\cos 2x = 0$	or	$\cos x = 0$
$5x = (2n+1)\frac{\pi}{2}$		$2x = (2n+1)\frac{\pi}{2}$		
$x = (2n+1)\frac{\pi}{10}$		$x = (2n+1)\frac{\pi}{4}$		$x = (2n+1)\frac{\pi}{2}$
$(n \in I)$		$(n \in I)$		$(n \in I)$

Q.5 Solve: $\operatorname{cosec} x - \operatorname{cosec} 2x = \operatorname{cosec} 4x$

A. $\frac{1}{\sin x} - \frac{1}{\sin 2x} = \frac{1}{\sin 4x}$

$\Rightarrow \frac{1}{\sin x} = \frac{1}{\sin 2x} + \frac{1}{\sin 4x} = \frac{\sin 2x + \sin 4x}{\sin 2x.\sin 4x}$

$\Rightarrow \frac{1}{\sin x} = \frac{2\sin 3x.\cos x}{\sin 2x.\sin 4x}$

$\Rightarrow \sin 2x.\sin 4x = \sin 3x\ (2\sin x \cos x)$

$\Rightarrow \sin 4x - \sin 3x = 0$

$\Rightarrow 2\cos\frac{7x}{2}\ \sin\frac{x}{2} = 0$	or	$\sin\frac{x}{2} = 0$
$\Rightarrow \cos\frac{7x}{2} = 0$	or	$\Rightarrow x/2 = n\pi$
$\Rightarrow x = (2n+1)\frac{\pi}{7}$		$\Rightarrow x = 2n\pi$ [Rejected]
$(n, m \in I, n \neq 7m - 4)$		(as $\sin x = 0$)

Q.6 Solve: $\frac{\sin 6x}{\sin x} = 8\cos x.\cos 2x.\cos 4x$

A. $\frac{\sin 6x}{\sin x} = 8.\frac{\sin 8x}{8\sin x}$

$\Rightarrow \sin 6x = \sin 8x$

$\Rightarrow \sin 8x - \sin 6x = 0$

$\Rightarrow 2\cos 7x.\sin x = 0$

$\Rightarrow \cos 7x = 0$ (as sinx cannot be zero)

$x = (2n+1)\frac{\pi}{14}$; $n \in I$

(v) Type – 5

Solving equations by change of variable or by substitution method.

Q.1 Solve: $\sin^4 2x + \cos^4 2x = \sin 2x.\cos 2x$

A. $(\sin^2 2x + \cos^2 2x)^2 - 2\sin^2 2x.\cos^2 2x = \frac{1}{2}\sin 4x$

$\Rightarrow 1 - \frac{1}{2}(2\sin 2x\cos 2x)^2 = \frac{1}{2}\sin 4x$

$\Rightarrow 2 - \sin^2 4x = \sin 4x$

$\Rightarrow \sin^2 4x + \sin 4x - 2 = 0$

$\Rightarrow (\sin 4x - 1)(\sin 4x + 2) = 0$

$\Rightarrow \sin 4x = 1$ or $\sin 4x = -2$

$4x = 2n\pi + \frac{\pi}{2}$ (Not possible)

$x = \frac{n\pi}{2} + \frac{\pi}{8} (n \in I)$

Previous Year's Questions

The number of soluitons of the equation $1 + \sin^4 x = \cos^2 3x$, $x \in \left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right]$ is

(A) 5 (B) 4

(C) 7 (D) 3

(vi) Type – 6

Solving equations with the use of the boundness of the function sinx, cosx

Q.1 Solve: $\sin^3x - \cos^3x = 1 + \sin x.\cos x$

A. $(\sin x - \cos x)(1 + \sin x \cos x) = 1 + \sin x.\cos x$

$\Rightarrow 1 + \sin x \cos x = 0$ or $\sin x - \cos x = 1$

$\Rightarrow 2\sin x\cos x = -2$ $\cos\left(x + \frac{\pi}{4}\right) = \cos\frac{3\pi}{4}$

$\Rightarrow \sin 2x = -2$ $x + \frac{\pi}{4} = 2n\pi \pm \frac{3\pi}{4}$

Not possible $x = 2n\pi + \frac{\pi}{2}, (2n-1)\pi \ (n \in I)$

Q.2 Solve: $\cos x + \cos 2x + \cos 3x = 3$

A. $(\cos x + \cos 3x) + \cos 2x = 3$

$\Rightarrow (2\cos 2x.\cos x) + \cos 2x = 3$

$\Rightarrow (\cos 2x)(2\cos x + 1) = 3$

$\Rightarrow 2\cos x + 1 = 3\sec 2x$

Now LHS $\in [-1,3]$

RHS $\in (-\infty,-3] \cup [3,\infty)$

$\Rightarrow$ LHS = RHS = 3

$\Rightarrow \cos x = 1$ & $\sec 2x = 1$

$x = 2n\pi$ $(n \in I)$ $\cos 2x = 1$ $x = m\pi (m \in I)$

$(x = 2n\pi)(n \in I)$

Alternate method:

Given equation only possible if

$\cos x = 1$ and $\cos 2x = 1$ and $\cos 3x = 1$

$x = 2n\pi$ and $x = n\pi$ and $x = \frac{2n\pi}{3}$

common values $x = 2n\pi, (n \in I)$

Q.3 If $0 \le x \le 2\pi$, then the number of real values of x, which satsify the equation $\cos x + \cos 2x + \cos 3x + \cos 4x = 0$

(A) 3 (B) 5 (C) 7 (D) 9

A. $(\cos x + \cos 4x) + (\cos 2x + \cos 3x) = 0$

$\Rightarrow 2\cos\frac{5x}{2}.\cos\frac{3x}{2} + 2\cos\frac{5x}{2}.\cos\frac{x}{2} = 0$

$\Rightarrow \quad 2\cos\frac{5x}{2}\left\{\cos\frac{3x}{2}+\cos\frac{x}{2}\right\}=0$

$\Rightarrow \quad \cos\frac{5x}{2}=0\,,\ \cos x=0,\ \cos\frac{x}{2}=0$

$\Rightarrow \quad x=(2n+1)\frac{\pi}{5}, x=(2n+1)\frac{\pi}{2}, x=(2n+1)\pi \quad (n\in I)$

Values in $[0, 2\pi]$ are

$x=\frac{\pi}{5},\frac{3\pi}{5},\pi,\frac{7\pi}{5},\frac{9\pi}{5},\frac{\pi}{2},\frac{3\pi}{2}$

Q.4 Solve: $\sin x(\cos\frac{x}{4}-2\sin x)+(\sin\frac{x}{4}-2\cos x)\cos x+1=0$

A. $(\sin x\cos\frac{x}{4}+\cos x\sin\frac{x}{4})-2(\sin^2x+\cos^2x)+1=0$

$\Rightarrow \quad \sin\frac{5x}{4}-2+1=0$

$\Rightarrow \quad \sin\left(\frac{5x}{4}\right)=1$

$\Rightarrow \quad \frac{5x}{4}=2n\pi+\pi/2$

$\Rightarrow \quad x=\frac{8}{5}n\pi+\frac{2\pi}{5} \quad (n\in I)$

Q.5 Solve for x and y: $x^2+2x+3=\sqrt{3}\ \sin y+\cos y$

A. $(x^2+2x+1)+2=2\left(\frac{\sqrt{3}}{2}\sin y+\frac{1}{2}\cos y\right)$

$\Rightarrow \quad (x+1)^2+2=2\sin\left(y+\frac{\pi}{6}\right)$

Now LHS $\in [2,\infty)$

RHS $\in [-2,2]$

Hence LHS = RHS = 2

$\mathbf{x=-1}$ and $\sin\left(y+\frac{\pi}{6}\right)=1$

$y+\frac{\pi}{6}=2n\pi+\frac{\pi}{2}$

$\mathbf{y=2n\pi+\frac{\pi}{3}} \quad (n\in I)$

Q.6 Solve x and y : $1 - 2x - x^2 = \tan^2(x + y) + \cot^2(x + y)$

A. $-(x + 1)^2 + 2 = \tan^2(x + y) + \dfrac{1}{\tan^2(x + y)}$

Now LHS $\in (-\infty, 2]$

RHS $\in [2, \infty)$

Hence LHS = RHS = 2

$x = -1$ and $\tan^2(x + y) = 1 = \tan^2\dfrac{\pi}{4}$

$\Rightarrow x + y = n\pi \pm \dfrac{\pi}{4}$

$-1 + y = n\pi \pm \dfrac{\pi}{4}$

$\mathbf{x = -1}$ and $\mathbf{y = n\pi \pm \dfrac{\pi}{4} + 1}$ $(n \in I)$

Q.7 Number of ordered pair satisfying the inequality $2^{\text{cosec}^2 x}\sqrt{y^2 - 2y + 2} \le 2$

A. $\because\ 2^{\text{cosec}^2 x} \in [2, \infty)$

and $\sqrt{y^2 - 2y + 2} = \sqrt{(y-1)^2 + 1} \in [1, \infty)$

Hence $2^{\text{cosec}^2 x}\sqrt{y^2 - 2y + 2} \in [2, \infty)$

$\therefore$ given inequality will only be satisfy if

$2^{\text{cosec}^2 x}\sqrt{y^2 - 2y + 2} = 2$

$\Rightarrow \text{cosec}^2 x = 1$ and $y^2 - 2y + 2 = 1$

$\sin^2 x = \sin^2\dfrac{\pi}{2}$ and $(y - 1)^2 = 0$

$\mathbf{x = n\pi \pm \dfrac{\pi}{2}}$ $(n \in I)$ and $\mathbf{y = 1}$

Q.8 Solve: $(\cos x)^{100} - (\sin x)^{100} = 1$

A. We know $\sin^{100} x, \cos^{100} x \in [0,1]$

Hence for above equation to satisfy.

$(\cos x)^{100} = 1$ and $(\sin x)^{100} = 0$

$\cos x = \pm 1$ and $\sin x = 0$

$x = n\pi (n \in I)$ and $x = n\pi (n \in I)$

$\cap$

$\mathbf{x = n\pi}\ (n \in I)$

(vii) Type – 7

Solution of trigonometric equations of the form of $f(x) = \sqrt{\phi(x)}$

Q.1 Solve: $\sqrt{1-\cos x} = \sin x$

A. On squaring

$(1 - \cos x) = \sin^2 x$

$\Rightarrow (1 - \cos x) = (1 - \cos x)(1 + \cos x)$

$\Rightarrow \cos x = 1$ or $1 + \cos x = 1$

$\mathbf{x = 2n\pi}$ $\cos x = 0$

$(n \in I)$ $\mathbf{x = 2n\pi + \frac{\pi}{2}}$ $(n \in I)$

(as $\sin x \geq 0$)

Q.2 Solve: $2\sin\left(3x + \frac{\pi}{4}\right) = \sqrt{1 + 8\sin 2x \cdot \cos^2 2x}$

A. $2\left(\frac{1}{\sqrt{2}}\sin 3x + \frac{1}{\sqrt{2}}\cos 3x\right) = \sqrt{1 + 4\sin 4x \cdot \cos 2x}$

On squaring

$2(\sin 3x + \cos 3x)^2 = 1 + 2(2\sin 4x \cos 2x)$

$2(1 + \sin 6x) = 1 + 2(\sin 6x + \sin 2x)$

$2 + 2\sin 6x = 1 + 2\sin 6x + 2\sin 2x$

$\sin 2x = \frac{1}{2} = \sin\frac{\pi}{6}$

$2x = n\pi + (-1)^n \frac{\pi}{6};\ n \in I$

$x = \frac{n\pi}{2} + (-1)^n \frac{\pi}{12};\ n \in I$ ...(1)

Now we must keep in mind that $\sin\left(3x + \frac{\pi}{4}\right) > 0$ hence

let us check in (1) for

$n = 0,\ x = \frac{\pi}{12}$ $\Rightarrow \sin\left(3x + \frac{\pi}{4}\right) > 0$

$n = 1,\ x = \frac{5\pi}{12}$ $\Rightarrow \sin\left(3x + \frac{\pi}{4}\right) < 0$

$n = 2, \frac{13\pi}{12} \quad \Rightarrow \sin\left(3x + \frac{\pi}{4}\right) < 0$

$n = 3, \frac{17\pi}{12} \quad \Rightarrow \sin\left(3x + \frac{\pi}{4}\right) > 0$

$n = 4, \frac{25\pi}{12} \quad \Rightarrow \sin\left(3x + \frac{\pi}{4}\right) > 0$

⋮

Hence possible values of n

$n = 0, 3, 4, 7, 8, 11, \ldots$

for $0, 4, 8, \ldots$ $n = 4m$

$3, 7, 11, \ldots$ $n = 4m + 3$

hence in (1)

(i) Put $n = 4m$ then solution

$$\mathbf{x = 2m\pi + \frac{\pi}{12}} \ (m \in I)$$

(ii) Put $n = 4m + 3$ then solution

$$x = (4m + 3)\frac{\pi}{2} - \frac{\pi}{12} = 2m\pi + \frac{3\pi}{2} - \frac{\pi}{12}$$

$$x = 2m\pi + \frac{17\pi}{12}(m \in I)$$

Q.3 Find smallest positive x satisfying $\sqrt{1 - \sin 2x} - \sqrt{2}\sin 3x = 0$

A. $\sqrt{1 - \sin 2x} - \sqrt{2}\sin 3x = 0$

On squaring

$(1 - \sin 2x) = 2\sin^2 3x = 1 - \cos 6x$

$\Rightarrow \sin 2x = \cos 6x$

$\Rightarrow \cos 6x = \cos\left(\frac{\pi}{2} - 2x\right)$

$\Rightarrow 6x = 2n\pi \pm \left(\frac{\pi}{2} - 2x\right)$

$\Rightarrow 8x = 2n\pi + \frac{\pi}{2}, 4x = 2n\pi - \frac{\pi}{2}$

$$x = \frac{n\pi}{4} + \frac{\pi}{16}, \ x = \frac{n\pi}{2} - \frac{\pi}{8}(n \in I)$$

Smallest positive value is $\mathbf{\frac{\pi}{16}}$

(viii) Type – 8 (Miscellenous)

Q.1 Find the least positive angle measure in degree satisfying the equation $\sin^3x + \sin^32x + \sin^33x = (\sin x + \sin 2x + \sin 3x)^3$

A. Let $\sin x = a$, $\sin 2x = b$, $\sin 3x = c$

Hence given equation is

$a^3 + b^3 + c^3 = (a + b + c)^3$

$\Rightarrow a^3 + b^3 + c^3 = a^3 + b^3 + c^3 + 3(a + b)(b + c)(c + a)$

$\Rightarrow a + b = 0\ ,\ b + c = 0\ ,\ c + a = 0$

$$2\sin\frac{3x}{2}\cos\frac{x}{2} = 0,\ \ 2\sin\frac{5x}{2}\cos\frac{x}{2} = 0\ ,\ 2\sin 2x\cos x = 0$$

Least positive angle is $x = \mathbf{72°}$

Q.2 Let S be the set of all $\alpha \in R$ such that the equation, $\cos 2x + \alpha\sin x = 2\alpha - 7$ has a solution. Then S is equal to

(A) [2,6] (B) [3,7] (C) R (D) [1,4]

A. Given equation is

$1 - 2\sin^2x + \alpha\sin x = 2\alpha - 7$

$\Rightarrow 2\sin^2x - \alpha\sin x + 2(\alpha - 4) = 0$

$$\sin x = \frac{\alpha \pm \sqrt{\alpha^2 - 4.2.2(\alpha - 4)}}{4}$$

$$\sin x = \frac{\alpha \pm \sqrt{\alpha^2 - 16\alpha + 64}}{4}$$

$$\sin x = \frac{\alpha \pm (\alpha - 8)}{4} = \frac{\alpha - 4}{2}, 2$$

Equation have a solutions if $-1 \le \frac{\alpha - 4}{2} \le 1 \Rightarrow \mathbf{\alpha \in [2,6]}$

Q.3 If $\tan(5\pi\cos\alpha) = \cot(5\pi\sin\alpha)$ number of solutions in $[0, 2\pi]$

A. $$\tan(5\pi\cos\alpha) = \tan\left(\frac{\pi}{2} - 5\pi\sin\alpha\right)$$

$$\Rightarrow 5\pi\cos\alpha = n\pi + \frac{\pi}{2} - 5\pi\sin\alpha\ \ (n \in I)$$

$$\Rightarrow 5\pi(\sin\alpha + \cos\alpha) = n\pi + \frac{\pi}{2}$$

$\Rightarrow \sin\alpha + \cos\alpha = \frac{n}{5} + \frac{1}{10} = \frac{2n+1}{10}$ ... (1)

$\because \quad -\sqrt{2} \le \sin\alpha + \cos\alpha \le \sqrt{2}$

$-\sqrt{2} \le \frac{2n+1}{10} \le \sqrt{2}$

$\Rightarrow -\frac{10\sqrt{2}-1}{2} \le n \le \frac{10\sqrt{2}-1}{2}$

$\Rightarrow n = -7, -6, -5, ..., 6$

for each n, equation (1) have two values of α in $[0,2\pi]$ hence total solutions.

$= 14 \times 2 = \mathbf{28}$

Q.4 Solve: $\cos3x - \cos2x = \sin3x$

A. $\cos3x - \sin3x = \cos2x$

$(4\cos^3x - 3\cos x) - (3\sin x - 4\sin^3x) = \cos^2x - \sin^2x$

$\Rightarrow 4(\sin^3x + \cos^3x) - 3(\sin x + \cos x) = (\cos x + \sin x)(\cos x - \sin x)$

$\Rightarrow (\sin x + \cos x)\{4(1 - \sin x\cos x) - 3 - (\cos x - \sin x)\} = 0$

$\Rightarrow (\sin x + \cos x)\{1 - 4\sin x\cos x - \cos x + \sin x\} = 0$

$\Rightarrow \sin x + \cos x = 0 \Rightarrow \tan x = -1$

$x = n\pi - \frac{\pi}{4} (n \in I)$

or $2(1 - 2\sin x\cos x) + (\sin x - \cos x) - 1 = 0$

$2(\sin x - \cos x)^2 + (\sin x - \cos x) - 1 = 0$

Let $\sin x - \cos x = \alpha$, then

$\Rightarrow 2\alpha^2 + \alpha - 1 = 0$

$\Rightarrow (\alpha + 1)(2\alpha - 1) = 0$

$\Rightarrow \alpha = -1$ or $\alpha = \frac{1}{2}$

$\sin x - \cos x = -1$ $\qquad$ $\sin x - \cos x = \frac{1}{2}$

$\sin\left(x - \frac{\pi}{4}\right) = \frac{-1}{\sqrt{2}}$ $\qquad$ $\sin\left(x - \frac{\pi}{4}\right) = \frac{1}{2\sqrt{2}}$

$x - \frac{\pi}{4} = n\pi - (-1)^n \frac{\pi}{4}$ $\qquad$ $x - \frac{\pi}{4} = n\pi + (-1)^n \alpha$

$(n \in I)$ $\qquad$ $\mathbf{(n \in I), \sin\alpha = \frac{1}{2\sqrt{2}}}$

Previous Year's Questions

If $0 \le x \le \frac{\pi}{2}$, then the number of value of x for which $\sin x - \sin 2x + \sin 3x = 0$, is

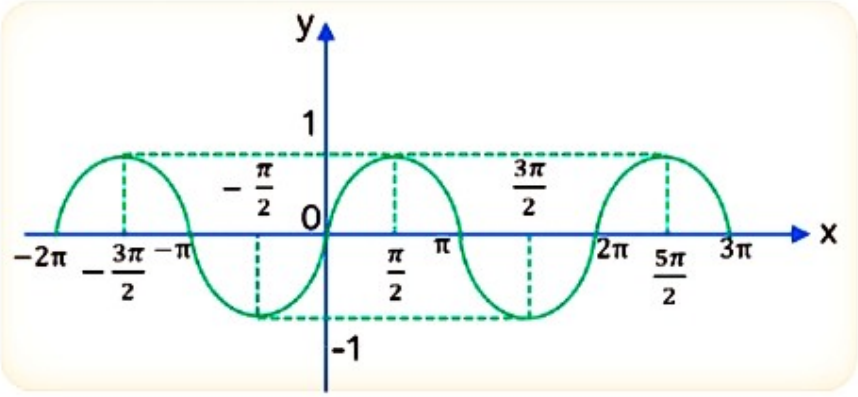
(A) 2 (B) 1

(C) 3 (D) 4

TRIGONOMETRIC INEQUALITIES AND SYSTEM OF INEQUALITIES

Q.1 Solve the inequality $\sin x > 0$

A.



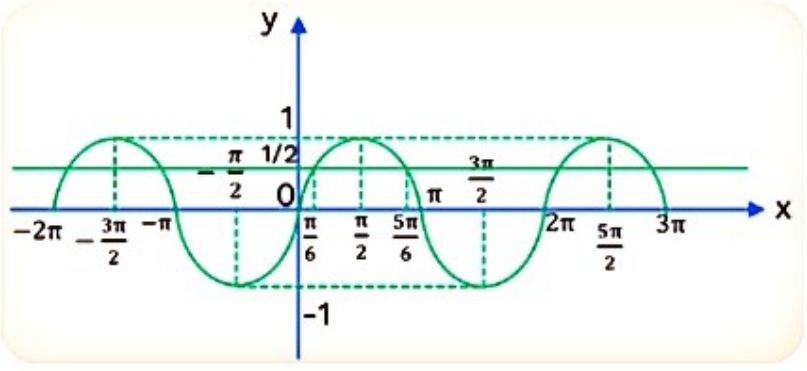
$$x \in \ldots(-2\pi, -\pi) \cup (0, \pi) \cup (2\pi, 3\pi)$$

General solution

$$x \in \bigcup_{n\in I} (2n\pi, (2n+1)\pi)$$

Q.2 Solve the inequality $\log_2\left(\sin\frac{x}{2}\right) < -1$

A. $0 < \sin\frac{x}{2} < \frac{1}{2}$



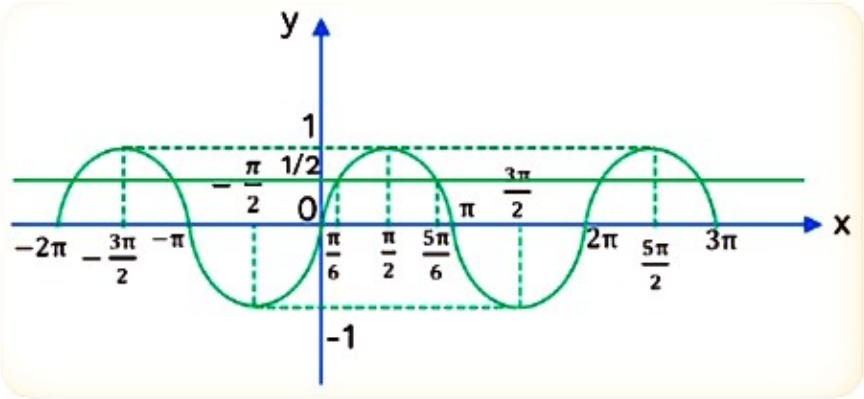
General solution

$$\frac{x}{2} \in \bigcup_{n\in I}\left(2n\pi, 2n\pi + \frac{\pi}{6}\right) \cup \left(2n\pi + \frac{5\pi}{6}, (2n+1)\pi\right)$$

$$x \in \bigcup_{n\in I}\left(4n\pi, 4n\pi + \frac{\pi}{3}\right) \cup \left(4n\pi + \frac{5\pi}{3}, (4n+2)\pi\right)$$

Q.3 Solve the inequality $\log_2(\sin x) < -1$

A. $0 < \sin x < \frac{1}{2}$

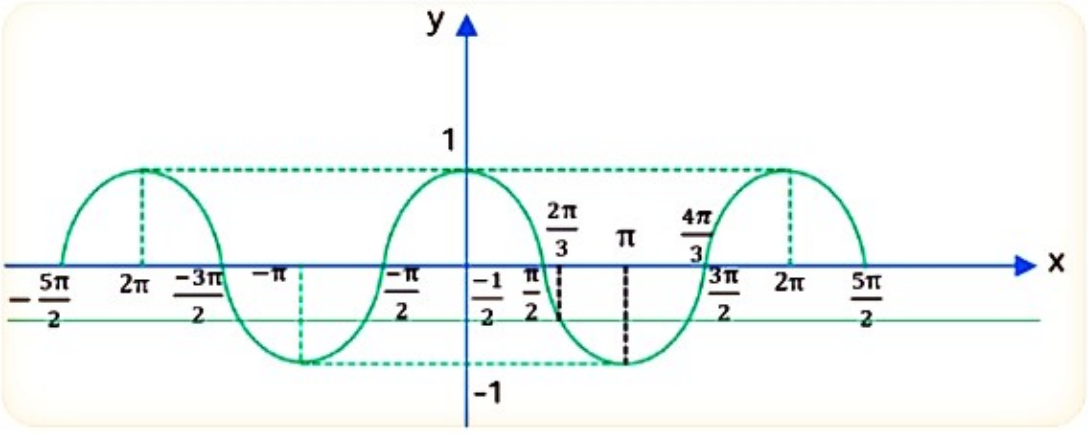


General solution

$$x \in \bigcup_{n\in I}\left(2n\pi, 2n\pi + \frac{\pi}{6}\right) \cup \left(2n\pi + \frac{5\pi}{6}, (2n+1)\pi\right)$$

Q.4 Solve the inequality $\cos x < -\frac{1}{2}$

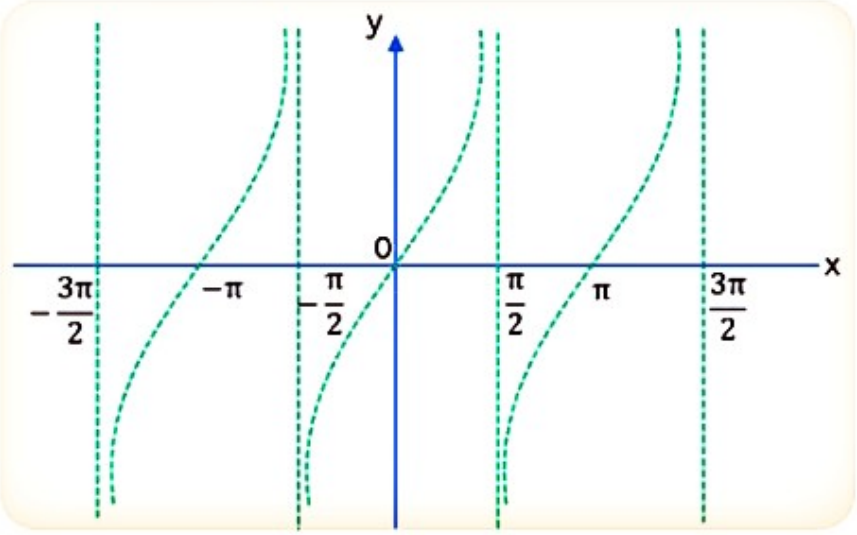
A.



General solution: $x \in \bigcup_{n\in I}\left(2n\pi + \frac{2\pi}{3}, 2n\pi + \frac{4\pi}{3}\right)$

Q.5 Solve the inequality $\tan x > 0$

A. $x \in \ldots \left(-\pi, -\frac{\pi}{2}\right) \cup \left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$



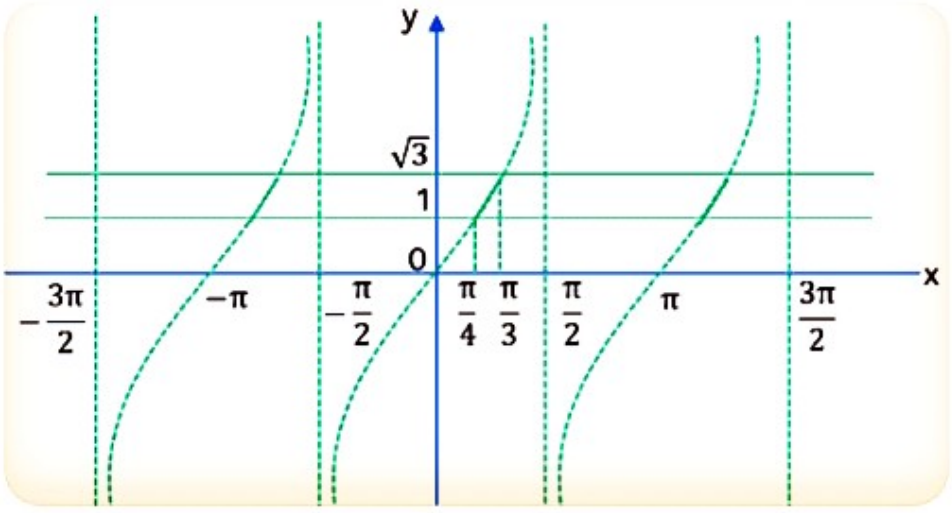
General solutions: $x \in \bigcup_{n \in I} \left(n\pi, n\pi + \frac{\pi}{2}\right)$

Q.6 Solve the inequality $\tan^2 x - \left(\sqrt{3}+1\right)\tan x + \sqrt{3} < 0$

A. $(\tan x - 1)\left(\tan x - \sqrt{3}\right) < 0$

$\Rightarrow \quad 1 < \tan x < \sqrt{3}$

General solution: $x \in \bigcup_{n \in I} \left(n\pi + \frac{\pi}{4}, n\pi + \frac{\pi}{3}\right)$



Q.7 Solve the inequality $\sin 2x > \sqrt{2}\sin^2 x + \left(2-\sqrt{2}\right)\cos^2 x$

A. Divide by $\cos^2 x$

$$\frac{2\sin x\cos x}{\cos^2 x} > \sqrt{2}\frac{\sin^2 x}{\cos^2 x} + \left(2-\sqrt{2}\right)$$

$$\Rightarrow \quad 2\tan x > \sqrt{2}\tan^2 x + \left(2-\sqrt{2}\right)$$

$$\Rightarrow \quad \sqrt{2}\tan^2 x - \sqrt{2} + 2 - 2\tan x < 0$$

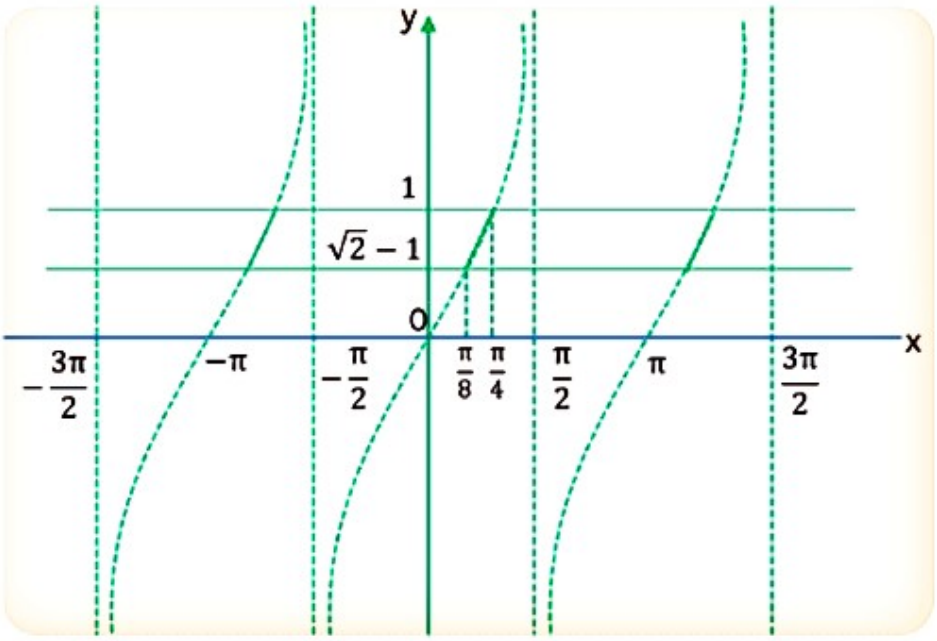
$$\Rightarrow \quad \sqrt{2}\left(\tan^2 x - 1\right) + 2\left(1 - \tan x\right) < 0$$

$$\Rightarrow \quad \sqrt{2}\left(\tan x - 1\right)\left\{\tan x + 1 - \sqrt{2}\right\} < 0$$

$$\Rightarrow \quad \sqrt{2} - 1 < \tan x < 1$$

General solutions

$$x \in \bigcup_{n\in} \left(n\pi + \frac{\pi}{8}, n\pi + \frac{\pi}{4}\right)$$



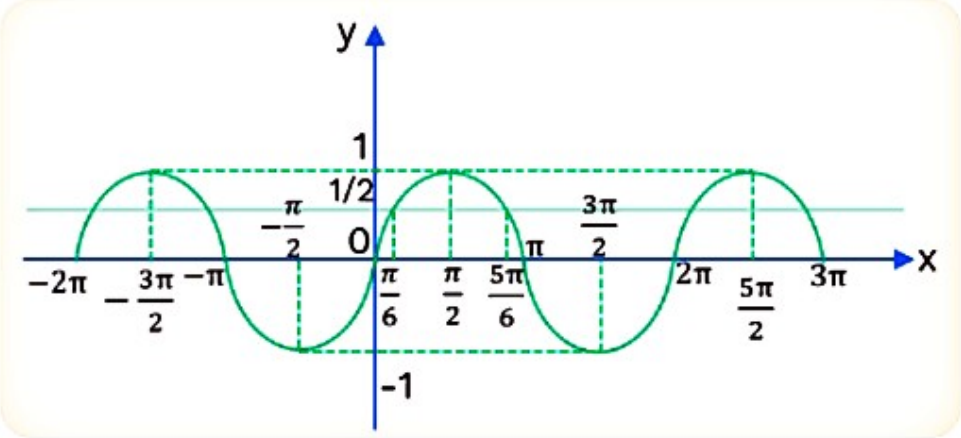
Q.8 Solve the inequality $\sin x \geq \cos 2x$.

A. $\sin x \geq 1 - 2\sin^2 x$

$$\Rightarrow \quad 2\sin^2 x + \sin x - 1 \geq 0$$

$$(2\sin x - 1)(\sin x + 1) \geq 0$$

$$\Rightarrow \quad \sin x \in \left[\frac{1}{2}, 1\right] \cup \{-1\}$$



General Solution:

$$x \in \bigcup_{n\in I}\left[2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6}\right] \cup \left\{2n\pi + \frac{3\pi}{2}\right\}$$

Q.9 Solve the inequality $\sqrt{5 - 2\sin x} \geq 6\sin x - 1$

A. Let $\sin x = t$ then

$\sqrt{5 - 2t} \geq 6t - 1$

I – case: $6t - 1 \geq 0 \Rightarrow t \geq \frac{1}{6}$

On squaring

$\Rightarrow 5 - 2t \geq (6t - 1)^2$

$\Rightarrow 5 - 2t \geq 36t^2 - 12t + 1$

$\Rightarrow 36t^2 - 10t - 4 \leq 0$

$\Rightarrow 18t^2 - 5t - 2 \leq 0$

$\Rightarrow 18t^2 - 9t + 4t - 2 \leq 0$

$\Rightarrow 9t(2t - 1) + 2(2t - 1) \leq 0$

$\Rightarrow (9t + 2)(2t - 1) \leq 0$

$$t \in \left[\frac{-2}{9}, \frac{1}{2}\right]$$

by intersection of all these we get

$$t \in \left[\frac{1}{6}, \frac{1}{2}\right] \quad \ldots (1)$$

II – Case: $6t - 1 < 0 \Rightarrow t < \frac{1}{6}$

$\Rightarrow$ positive $\geq$ negative

always true

$t \in R$

By Intersection of all case II we get $t < \frac{1}{6}$

Previous Year's Questions

If $0 < \theta < 2\pi$, then the intervals of value of q for which $2\sin^2\theta - 5\sin\theta + 2 > 0$, is

(A) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right)$

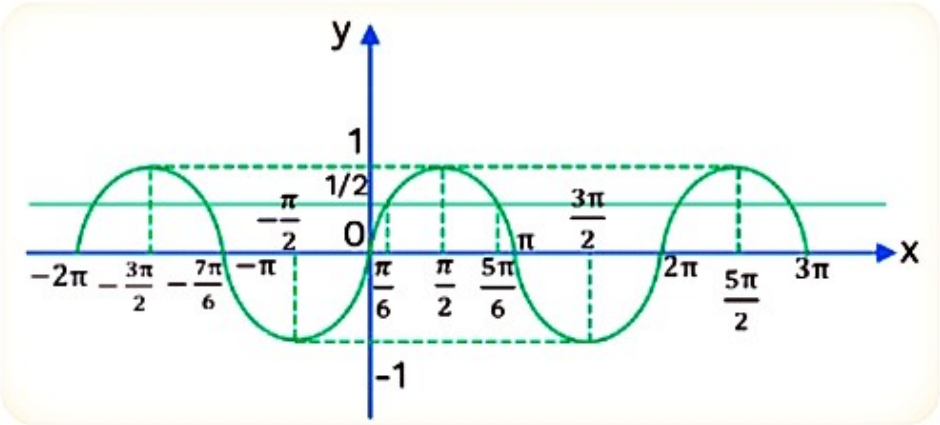
(B) $\left(\frac{\pi}{8}, \frac{5\pi}{6}\right)$

(C) $\left(0, \frac{\pi}{8}\right) \cup \left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$

(D) $\left(\frac{41\pi}{48}, \pi\right)$

(1) ∪ (2)

$$t \le \frac{1}{2} \Rightarrow \sin x \le \frac{1}{2}$$



$$x \in \bigcup_{n \in I}\left[2n\pi - \frac{7\pi}{6}, 2n\pi + \frac{\pi}{6}\right]$$

System of Ttrigonometric Equations

Q.1 $\cos x \cdot \cos y = \frac{3}{4}$ and $\sin x \sin y = \frac{1}{4}$. Find general solution of x and y.

A. On adding $\cos x \cos y + \sin x \sin y = 1$

$\cos(x - y) = 1 \Rightarrow x - y = 2n\pi \ (n \in I)$ ...(1)

On subtracting

$$\cos x \cos y - \sin x \sin y = \frac{1}{2}$$

$$\cos(x + y) = \frac{1}{2}$$

$$x + y = 2m\pi \pm \frac{\pi}{3} \quad \text{...(2)}$$

(1) + (2) $\Rightarrow x = n\pi + m\pi \pm \frac{\pi}{6} \Rightarrow x = k\pi \pm \frac{\pi}{6} \ (k \in I)$

(2) − (1) $\Rightarrow y = m\pi - n\pi \pm \frac{\pi}{6}$

$\Rightarrow y = p\pi \pm \frac{\pi}{6} (p \in I)$

Q.2 $\cos(x+y) = 1$ and $\cos(x-y) = \frac{1}{2}$. Find general solution of x and y

A. $\cos(x+y) = 1$

$\Rightarrow x + y = 2m\pi \ (m \in I)$ ... (1)

$\cos(x-y) = \frac{1}{2}$

$\Rightarrow x - y = 2n\pi \pm \frac{\pi}{3} (n \in I)$ ... (2)

(1) + (2)

$\Rightarrow x = m\pi + n\pi \pm \frac{\pi}{6} \Rightarrow x = k\pi \pm \frac{\pi}{6} (k \in I)$

(1) – (2)

$\Rightarrow y = m\pi - n\pi \mp \frac{\pi}{6} \Rightarrow y = p\pi \pm \frac{\pi}{6} (p \in I)$

Q.3 Find real values of x for which $27^{\cos 2x} . 81^{\sin 2x}$ is minimum, also find this minimum value.

A. Given expression becomes,

$\Rightarrow 3^{3\cos 2x} . 3^{4\sin 2x} = 3^{3\cos 2x + 4\sin 2x}$

For minimum value

$3\cos 2x + 4\sin 2x = -5$

divide by 5

$\Rightarrow \frac{3}{5}\cos 2x + \frac{4}{5}\sin 2x = -1$

$\Rightarrow \sin(2x + \alpha) = -1 \ ; \ \alpha = \sin^{-1}\left(\frac{3}{5}\right)$

$\Rightarrow 2x + \alpha = 2n\pi + \frac{\pi}{2} \Rightarrow x = n\pi + \frac{\pi}{4} - \frac{1}{2}\sin^{-1}\left(\frac{3}{5}\right) \ (n \in I)$

Minimum Value $= 3^{-5} = \frac{1}{3^5} = \frac{1}{243}$

Q.4 $\cos(\alpha-\beta)=1$ and $\cos(\alpha+\beta)=1/e$, where $\alpha,\beta\in[-\pi,\pi]$, numbers of pairs of α,β which satisfy both the equation is

(A) 0 (B) 1 (C) 2 (D) 4

A. $\because\ \alpha,\beta\in[-\pi,\pi]$

$\therefore\ \alpha+\beta\in[-2\pi,2\pi],\alpha-\beta\in[-2\pi,2\pi]$

Now $\cos(\alpha-\beta)=1\Rightarrow\alpha-\beta=-2\pi,0,2\pi$

$\left.\begin{matrix}\text{for }\alpha-\beta=-2\pi\Rightarrow\alpha=-\pi,\beta=\pi\\ \alpha-\beta=2\pi\Rightarrow\alpha=\pi,\beta=-\pi\end{matrix}\right\}\Rightarrow\cos(\alpha+\beta)=1$ So, rejected

for $\alpha-\beta=0\Rightarrow\alpha=\beta$

and $\cos(\alpha+\beta)=\cos2\alpha=\frac{1}{e}\rightarrow$ **4 solutions**

Q.5 If $0<\theta<2\pi$, then the intervals of values of θ for which $2\sin^2\theta-5\sin\theta+2>0$ is

(A) $\left(0,\frac{\pi}{6}\right)\cup\left(\frac{5\pi}{6}.2\pi\right)$ (B) $\left(\frac{\pi}{8},\frac{5\pi}{6}\right)$

(C) $\left(0,\frac{\pi}{8}\right)\cup\left(\frac{\pi}{6},\frac{5\pi}{6}\right)$ (D) $\left(\frac{41\pi}{48},\pi\right)$

A. **A**

Given inequality is $2\sin^2\theta-4\sin\theta-\sin\theta+2>0$

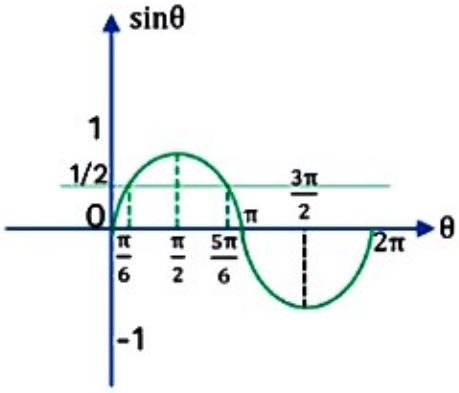
$\Rightarrow\ 2\sin\theta(\sin\theta-2)-(\sin\theta-2)>0$

$\Rightarrow\ (2\sin\theta-1)(\sin\theta-2)>0$

$\Rightarrow\ (2\sin\theta-1)<0\{\text{as}(\sin\theta-2)<0\}$

$\Rightarrow\ \sin\theta<\frac{1}{2}$

$x\in\left(0,\frac{\pi}{6}\right)\cup\left(\frac{5\pi}{6},2\pi\right)$



Q.6 Let $P = \{\theta : \sin\theta - \cos\theta = \sqrt{2}\cos\theta\}$ and

$Q = \{\theta : \sin\theta + \cos\theta = \sqrt{2}\sin\theta\}$ be two sets. then

(A) $P \subset Q$ and $Q - P \neq Q$ (B) $Q \not\subset P$

(C) $P \subset Q$ (D) $P = Q$

A. **D**

For P : $\sin\theta = (\sqrt{2}+1)\cos\theta$

$\Rightarrow \tan\theta = \sqrt{2}+1$ ... (1)

For Q : $\cos\theta = (\sqrt{2}-1)\sin\theta$

$\Rightarrow \tan\theta = \dfrac{1}{\sqrt{2}-1} = \sqrt{2}+1$ ... (2)

From (1) & (2) it is clear that P = Q